

NAME: _____

PERIOD: _____

DATE: _____

Two Interfaces, Twelve Specialists

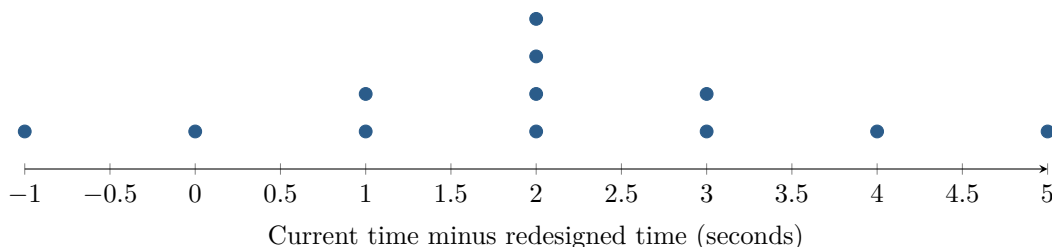
AP Statistics · Unit 4 · Topic 4.2 · B: 2027-01-14 · E: 2027-03-01

[STAR] TODAY'S PROBLEM

Suppose a hospital randomly samples 12 of its 180 billing specialists. Each completes the same task with the current interface and, one week later, the redesign; everyone uses the current interface first and there is no comparison group. Let d = current – redesigned time in seconds. The differences are shown.

Priya: “Use a matched-pairs t interval for μ_d ; fixed order limits a causal claim.”

Mateo: “Two columns require an independent two-sample interval; a positive interval proves the redesign caused faster work.”



FIRST TAKE — before the video (5 min)

Whose procedure is more defensible, Priya's or Mateo's? Cite how the two measurements are linked.

[BOOK] MATCH THE INTERVAL TO THE PAIRS (keep on the page)

For paired measurements, define every difference in the same order and treat the n pairs as one sample: $\bar{d} \pm t^*(s_d/\sqrt{n})$ with $df = n - 1$. Check conditions on the differences: random sampling or random assignment, $n \leq 0.10N$ when sampling without replacement, and, for $n < 30$, no strong skew or outliers. Two readings from each person are not two independent samples. Random sampling supports generalization to the sampled population; random assignment or an adequate randomized comparison is needed to isolate causation from practice or time effects.

Finish after the video:

DOK 1 (a) Give the number of observational units and pairs, define μ_d , state df , and interpret a positive difference.

DOK 2 (b) Check random sampling, 10 percent, and the shape of the differences. Compute $\bar{d}$, s_d , SE, and a 95 percent interval for μ_d ; interpret it.

DOK 3 (c) Choose the warranted analysis from the linked data. Use the interval for the 180 specialists, then use fixed order and no comparison group to bound causation. Explain the two consequences of the rejected procedure and causal wording.

Frame: A _____ interval is warranted because each d links _____, so the sample size is _____.

Frame: The interval supports _____ for the 180 specialists, but it does not prove _____ because _____.

EXIT — turn this in

One thing the video changed about my first take: _____